

COS30082 Applied Machine Learning



Lecture 11. RAG, Agentic AI, Multimodal ML

Topics

- Retrieval-Augmented Generation (RAG)
- Agentic AI - models that act, not just answer
- Multimodal ML - one model across text, image, audio, video and more

Retrieval-Augmented Generation (RAG)

Letting LLMs reason over data they were never trained on

Why RAG exists

- LLMs are powerful but have three structural gaps:
 - Knowledge cutoff - they do not know events after training
 - No access to private data - your PDFs, your customer databases, your codebase
 - Hallucination - they may make up information when they do not have enough evidence.
- Fine-tuning can teach new knowledge, but it is slow, expensive, hard to audit, and ages within weeks.
- RAG (Lewis et al., NeurIPS 2020) addresses this by injecting evidence into the prompt at query time, rather than encoding it in the model weights.

The RAG pipeline

Two phases:

- Indexing (offline)

Documents -> Chunks -> Embeddings -> Vector store

- Query (online)

User question → Embed query → Retrieve top-k relevant chunks from vector store → Build prompt with retrieved chunks → LLM generates answer

This way, the model sees the relevant evidence BEFORE it has a chance to make something up.

- An embedding is a dense vector that captures meaning.
 - Produced by a Transformer encoder
 - Cosine similarity in vector space $\sim$ semantic similarity in text
- Advantage over keyword-based search :
 - Semantic retrieval can identify conceptually relevant documents even when there is little or no lexical overlap.
 - For example, the query “How do I cancel my subscription?” may match the document “Account termination procedure” because both express the same underlying intent.

- A single vector cannot adequately represent all topics, details, and contexts in a long document.
- Key Design Choices in RAG
 - Chunk size: usually 200–800 tokens (one English word is about 1.3 tokens.)
 - Chunk overlap: repeat nearby text to keep context.
 - Chunk boundaries: split at natural text structures.
 - Metadata: add source details to each chunk, e.g. document name, page number, section title, date, and author.
 - Hierarchy: link chunks back to their parent section/document.

Vector search alone is often not enough. Real-world RAG systems usually combine several retrieval techniques.

- Hybrid search: semantic vectors + keyword search
 - User asks: “What is the late penalty for COS30082?”
 - RAG does:
 - Vector search finds chunks about assessment rules.
 - Keyword search finds the exact term “COS30082” and “late penalty”.
 - The system combines both results.
- Reranking: retrieve many chunks, then rerank the best ones
 - User asks: “When is the project due?”
 - RAG does:
 - First retrieves 50 possible chunks about assessments, schedule, and project.
 - A reranker checks which chunks best answer the question.
 - It keeps the top 5 most relevant chunks for the LLM.

Retrieval in Production RAG (continued)

- Query rewriting: clarify the user query before search
 - User asks: “When do we submit it?”
 - RAG does:
 - Rewrites the query as: “When is the COS30082 project submission due?”
 - Then it searches using the clearer question.
- Metadata filtering: search only within selected users, dates, or document types
 - User asks: “What was the policy in 2026?”
 - RAG does:
 - Searches only documents where year = 2026.
 - Or searches only document_type = policy.
 - This avoids using old or irrelevant documents.

Agentic AI

From models that answer to models that act

From chatbot to agent

- A chatbot is a model that turns text into text.
- An agent is a model that takes actions in the world and observes the results.
- The model becomes the controller.
- The intelligence is in deciding WHAT TO DO NEXT given partial information.

What Agentic AI Does

Agentic AI = LLM reasoning + tool use + workflow control

- LLM for thinking: understand, reason, plan, decide
- Tools for action: search, code, files, databases, APIs
- Agent for coordination: plan → act → observe → reflect → continue

What Agents Offer

- Planning: break complex tasks into steps
- Tool use: lets AI access data or external tools.
- Memory: keep useful context across steps
- Reflection: check results and retry if needed
- Automation: complete multi-step tasks

Multi-agent AI is an AI system where multiple specialised agents work together to complete a task.

For example, a multi-agent system where several agents cooperate to produce a research report.

- Planner Agent — breaks the task into steps
- Research Agent — searches and retrieves evidence
- Analysis Agent — summarises and checks findings
- Writer Agent — produces the final report

Where agents already work in production

- Coding agents
 - Claude Code, GitHub Copilot, Codex
- Browser / computer-use agents
 - Anthropic Computer Use, OpenAI Operator, Google Project Mariner
- Operations and support
 - Automated ticket triage, refunds, scheduling

Multimodal ML

One model, many modalities

- Multimodal ML combines multiple types of data in one system.
- It learns relationships across different data types to improve understanding and prediction.
- Common modalities include: text, images, audio, video, sensor data
- Multimodal AI is useful when one data type alone is incomplete, but combining modalities can improve context and reduce ambiguity.

Example

Image only: detect a skin lesion

Text only: patient symptoms

Multimodal ML: image + clinical notes → better diagnosis

Typical Multimodal AI Pipeline

Input modalities

(text, image, audio, video)



Feature extraction / embeddings

(CNN, Transformer, audio encoder, etc.)



Fusion

Combine information from multiple modalities



Prediction / generation

classification, captioning, Q&A, recommendation, report generation

Common Multimodal Applications

- Image captioning: image → text description
AI describes what is in a photo.
- Visual question answering: image + question → answer
Example: “How many cars are in this image?”
- Autonomous driving: cameras + sensors + maps → driving decisions
Example: detect pedestrians and plan route and movement.
- Medical diagnosis: scan/image + clinical notes → diagnosis support
Example: X-ray plus patient history → make a diagnosis
- Video understanding: video frames + audio + text → event understanding
Example: summarising a lecture video.